📄 AI-BFSI Internship Project Documentation

SL Project Proposal – Subscription Waste Detector

Cover Page

Title: SL Project Proposal **Project Name:** Subscription Waste Detector **Program:** AI Specialist BFSI Program **Document Purpose:** This proposal presents the project concept, objectives, scope, proposed solution, technology stack, implementation approach, expected outcomes, and project plan for the Subscription Waste Detector.

Executive Summary

The Subscription Waste Detector is an AI-assisted financial analysis application designed to help users identify recurring subscription expenses, visualize monthly subscription spending, monitor renewal dates, and receive intelligent cost-saving recommendations.

The project addresses a common financial challenge caused by the increasing reliance on subscription-based digital services. Rather than functioning as a complete personal finance management system, the application focuses specifically on subscription expense analysis and financial awareness.

The proposed solution combines **Streamlit, Python, Pandas, Plotly, and the Groq API** to create an interactive web application capable of analyzing uploaded expense data and generating AI-powered recommendations.

Project Background

Subscription-based services have become integral to modern digital life. Consumers subscribe to streaming platforms, cloud storage, productivity software, educational platforms, and AI tools. While convenient, recurring payments accumulate over time, making it difficult to monitor overall subscription expenditure.

Automatic renewals often result in users continuing to pay for inactive subscriptions. Existing expense tracking methods provide transaction history but do not specifically highlight recurring subscriptions or offer intelligent recommendations for optimization.

These challenges motivated the development of the Subscription Waste Detector.

Problem Statement

Modern consumers increasingly rely on multiple subscription-based services. However, recurring payments often remain unnoticed due to automatic renewals and limited visibility. Existing expense management methods provide insufficient support for identifying recurring subscriptions, analyzing subscription spending, and generating personalized recommendations.

As a result, users experience avoidable recurring expenditure and reduced awareness of their subscription commitments.

Project Objectives

- Develop an application capable of analyzing uploaded expense records.
- Detect recurring subscription payments automatically.
- Present subscription information through interactive visualizations.
- Display monthly subscription expenditure.
- Generate AI-powered cost-saving recommendations using the Groq API.
- Improve financial awareness regarding recurring subscription expenses.

Proposed Solution

The proposed solution is an AI-assisted web application that enables users to:

- Upload transaction records in CSV or Excel format.
- Manually enter expense information.
- Process data using Python and Pandas to identify recurring subscriptions.
- Display results through an interactive Streamlit dashboard with Plotly charts.
- Send analyzed subscription information to the Groq API for personalized recommendations.

The solution emphasizes simplicity, usability, and practical financial insights without unnecessary complexity.

Project Scope

Included Scope

- CSV expense file upload
- Excel expense file upload
- Manual expense entry
- Subscription detection
- Monthly subscription analysis
- Interactive dashboard
- Plotly visualizations
- Renewal reminders
- AI-powered recommendations via Groq API

Excluded Scope

The following are outside the scope of Version 1:

- User authentication

- Database integration
- Cloud synchronization
- Bank API integration
- Email/SMS notifications
- Investment management
- Loan management
- Tax calculation
- Multi-user collaboration

These may be considered for **future development**.

Technology Stack

Component	Technology
Frontend	Streamlit
Programming Language	Python
Data Processing	Pandas
Visualization	Plotly
Spreadsheet Support	OpenPyXL
Artificial Intelligence	Groq API
Development Environment	VS Code

Development Methodology

The project follows a structured software development lifecycle:

1. Problem identification
2. Requirement analysis
3. Solution design
4. Application development
5. Testing and validation
6. Documentation

This ensures systematic development and alignment between the identified problem and the implemented solution.

Expected Deliverables

- Functional Streamlit web application
- Subscription detection module
- Interactive dashboard
- Expense visualization reports
- Renewal reminder functionality
- AI-generated subscription recommendations
- Complete project documentation

Expected Benefits

For Users

- Better understanding of recurring subscription expenses
- Improved financial awareness
- Easier identification of unnecessary subscriptions
- AI-assisted cost optimization
- Reduced manual effort in analyzing transactions

For the Project

- Demonstrates practical AI integration
- Applies data analytics to personal finance
- Addresses a real-world financial problem
- Provides a scalable foundation for future improvements

Project Feasibility

Technical Feasibility

The required technologies are mature, open-source, and suitable for implementing the proposed solution.

Operational Feasibility

Users only need basic expense records in CSV or Excel format to use the application.

Economic Feasibility

The project leverages open-source tools and the Groq API, minimizing development costs while providing AI capabilities.

Potential Risks

Risk	Mitigation
Poor-quality input data	Validate uploaded files before processing
Inconsistent transaction formats	Apply preprocessing and data cleaning
AI response variability	Provide recommendations as guidance rather than financial advice
Missing subscription patterns	Improve recurring transaction detection logic where necessary

Future Enhancements

Potential improvements include:

- User authentication
- Database integration

- Cloud synchronization
- Email notifications
- Bank statement integration
- Multi-user support
- Advanced spending analytics
- Multi-currency support

These are intentionally excluded from the current implementation.

Conclusion

The Subscription Waste Detector is proposed as a practical AI-assisted financial analysis application that addresses the challenge of recurring subscription expenses. By combining transaction analysis, interactive visualization, renewal reminders, and AI-generated recommendations, the project provides a focused solution for improving financial awareness.